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LESSONS LEARNED FROM MANAGING EPC INTERFACES IN BROWNFIELD PROJECTS

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Introduction

Brownfield Engineering, Procurement, and Construction (EPC) projects—those executed within or tying into live facilities—demand careful interface management to avoid costly delays and safety risks [1]. Unlike greenfield builds, brownfield projects must integrate new systems into existing operations, often requiring tie-ins during tightly scheduled plant shutdowns [2][3]. Mismanaged interfaces (the points of interaction between systems, teams, or contractors) frequently lead to design conflicts, construction rework, and schedule overruns [4]. Effective interface coordination, on the other hand, helps ensure work is executed safely without unplanned disruption to ongoing processes [5]. This article examines these challenges through the lens of the 70 km Amaken Gas Trunkline Project—a fictionalized but realistic case study—and distills key lessons learned for managing EPC interfaces in brownfield projects.

Case Study Background: The Amaken Gas Trunkline Project

The Amaken Gas Trunkline Project was a 70 km natural gas pipeline development in the Middle East, undertaken from 2022 to 2024. It was operated by the fictitious Amaken National Gas Co. (ANGC), aiming to connect a new gas field in the Amaken region to the existing Zahran Gas Processing Facility. The project's scope included pipeline construction across desert terrain and brownfield tie-ins at Zahran—a live gas processing plant handling millions of cubic feet per day. A primary contractor was engaged for the pipeline, while a separate contractor managed modifications inside the Zahran facility, meaning multiple EPC teams had overlapping scopes. Critically, the final hookup of the new trunkline required a five-day plant shutdown window in mid-2024 to cut into the operating gas headers. With production losses estimated at over \$1 million per day during shutdown, success hinged on flawless interface coordination and timing. ANGC implemented a formal Interface Management Plan early in the project, recognizing that managing interfaces between the project team and ongoing operations would be a critical success factor [1][2].

Project Timeline and Interfaces: Initial engineering (FEED) for the pipeline began in 2021, with detailed design and procurement in 2022. Construction of the pipeline right-of-way started in early 2023, while brownfield works at Zahran (e.g. installing tie-in valves, instrumentation, and metering systems) were done in parallel. Key interfaces included (1) the pipeline-to-plant tie-in (mechanical interface connecting new piping to an existing gas inlet header), (2) electrical and control interfaces integrating new pipeline controls with the plant's control system, and (3) contractual interfaces between the pipeline EPC contractor and the plant modification contractor. Throughout execution, the project team tracked over 50 interface points using an interface register, each with designated responsible parties and due dates. These ranged from design information handovers (e.g. existing plant as-built data) to on-site physical tie-in points. The case highlights how interface challenges manifested – from permit delays to rework – and how they were addressed.

Challenges in Brownfield EPC Interface Coordination

Working within an operating gas facility presented unique interface coordination challenges. Several real-world issues emerged on the Amaken Project that are common in brownfield construction:

1. Permit Delays and Live Facility Work Approvals

Gaining timely work permits in a live plant proved difficult. All hot work, excavations, and tie-in activities at Zahran required formal Permit-to-Work approval from operations. Early in the project, interface documentation gaps (such as missing method statements or risk assessments for tie-ins) led the plant operations team to delay issuing permits, pushing the schedule. This highlighted the need for close operations involvement and complete interface packages. In brownfield environments, coordinating with plant operations is paramount—effective management of the interface between the project team and the operations team is a critical success factor, especially when shutdowns or tie-ins are involved [1]. At Amaken, daily interface meetings were instituted between the EPC contractors and operations to expedite permit reviews and ensure all safety precautions (lock-outs, isolations, etc.) were in place. Safety and regulatory compliance are non-negotiable in live facilities, and proper interface planning helps maintain compliance while reducing the risk of incidents [6]. The lesson was clear: involve operations representatives early and integrate their requirements into the interface plan to prevent permit-related delays.

2. Overlapping Contractors and Scope Gaps

With separate contractors for the pipeline and the plant modifications, interface handoffs were a constant source of risk. For example, the pipeline contractor was responsible for installing a block valve station near the plant boundary, while the plant contractor would tie that into the facility. Misalignment in their schedules and expectations led to a coordination scramble: the plant team was ready for the tie-in work during the shutdown, but earlier pipeline welds were delayed awaiting a permit, threatening the shutdown timeline. Such scenarios exemplify how large EPC projects involve multiple contractors and client teams, and managing their interactions is complex without structured processes [7]. On Amaken, critical milestones had to be synchronized between contractors—a challenge also noted in other projects requiring field sequencing of multiple contractors [8]. To mitigate this, the project employed an Interface Responsibility Matrix (IRM) that clearly delineated each contractor's scope at interface points and held joint coordination workshops. One important interface was scheduling shutdown tasks: both contractors had to finish certain preparatory works before the plant outage. Regular interface coordination meetings (weekly and then daily approaching the shutdown) were held to update status and ensure no task “fell through the cracks.” The project learned that formal communication protocols and defined responsibilities are crucial to prevent scope gaps between overlapping teams [9][10]. Without such clarity, ambiguities can lead to “I thought you were doing that” scenarios, causing delays.

3. Engineering Revision Cycles and Late Design Changes

Brownfield projects often face evolving designs as new information arises from the existing facility. The Amaken project was no exception—multiple engineering revision cycles occurred for the tie-in design. Initially, the new trunkline was to tie into an existing 24" header at the plant. During detailed design, the team discovered through site measurement that the header's wall thickness was greater than on the outdated drawings, prompting a recalculation of weld procedures and a change in the tie-in fitting design. Additionally, late in the project, operations requested an extra isolation valve and a bypass line for flexibility, forcing a design revision just weeks before the shutdown.

These changes cascaded through procurement and construction, illustrating how modifications in design or scope affect multiple interfaces and, without proper change management, can create conflicts and delays [11]. [12]. In one notable industry case, a design change for equipment tie-ins that was not properly communicated led to missing components and significant rework [13]. [14]. The Amaken team sought to avoid such missteps by using an Interface Change Control process: any change related to an interface (e.g. a tie-in detail or an I/O signal list update) triggered an interface review. This meant the interface register was updated and all stakeholders were notified of the change, as recommended by interface management best practices [15]. Despite this, the late engineering changes compressed the schedule, requiring out-of-sequence work and extra resources. The lesson learned is that early interface identification and freeze is ideal, but when changes are unavoidable, a formal change control and communication process is vital to keep all parties aligned [16].

4. Construction Rework and Field Surprises

Even with thorough planning, the project encountered field surprises that led to rework. Upon excavation at one tie-in location, the crew found an undocumented underground water line crossing the pipeline route near the plant. Work had to stop until it was rerouted, causing rework and delay. In another instance, a prefabricated spool piece for the plant tie-in did not fit because an existing pipe flange was 15 cm off the assumed position—a result of old as-built drawings being out of date. These scenarios reinforce that surprises during brownfield construction can significantly impact schedule and cost; without proper planning, such risks result in rework and delays [17]. The project team had tried to mitigate this by performing site surveys and as-built verification early, but some discrepancies still slipped through. Industry best practices urge doing comprehensive site validation—including 3D laser scanning or physical measurements—to ensure design tie-ins match field reality [18][19]. For example, point cloud laser scanning was used in parts of the Amaken project to map the existing pipe rack, which helped identify interferences between the new pipeline supports and existing cables (preventing what could have been major rework) [20][21]. However, not all areas were scanned due to time/cost, and those became the Achilles' heel where rework occurred. The clear lesson: invest in as-built verification and site surveys for all critical interfaces, and involve plant maintenance personnel who know the facility's hidden issues. Catching discrepancies on paper is far cheaper than fixes in the field.

5. High-Stakes Shutdown Window and Tie-In Execution

Perhaps the most intense challenge was executing the pipeline's final tie-in within the high-stakes 5-day shutdown window. This shutdown was the only time the plant could safely connect the new line, align control systems, and test without gas flow—any overrun would mean extended lost production and erode trust with the operator. Such tie-in shutdowns are common in brownfield projects and require meticulous planning. Most brownfield projects require a shutdown or tie-in to existing facilities [22], and often the window is limited by production needs. At Amaken, every task during the outage was scripted in an hour-by-hour plan. Still, unforeseen issues can threaten the schedule—for example, a permit delay on day 1 or a faulty gasket can cascade into missed deadlines. In our case, a leaking valve was discovered during the tie-in, consuming precious repair time. This underscores why interface readiness reviews before shutdown are essential.

The project team conducted a “Tie-In Readiness Review” one month prior, verifying that all materials, spools, tools, and procedures for each interface were prepared and contingency plans were in place. Interfaces were also classified by whether they could be done live or required the shutdown, with an aim to minimize work in the outage window. By discussing options with the plant operator, some tie-ins were relocated to lines that could be isolated during normal operation, reducing the number of shutdown scope items [23]. In industry practice, such detailed tie-in planning (sometimes documented in a Tie-In & Shutdown Philosophy) and schedules for each discipline help reduce downtime [24][25]. The Amaken project’s rigorous approach paid off—despite minor hiccups, the trunkline was connected and tested within the 5-day window, allowing the plant to resume on schedule. The key takeaway is that alignment of interface work with shutdown schedules and thorough readiness checks can make the difference between success and failure in brownfield tie-ins [3][26].

Interface Management Strategies and Solutions Implemented

To address the above challenges, the Amaken Project employed several interface management strategies from day one. These strategies illustrate practical measures to control interfaces in brownfield EPC projects:

- **Early Interface Identification and Classification:** During project planning, the team identified all points where different parties or systems would interact—mechanical tie-ins, electrical connections, instrumentation signals, civil works interfaces, and even organizational interfaces (e.g., handovers between the design team and construction team). Each interface was categorized (technical, contractual, operational, etc.) and assigned an owner [27][28]. For example, the pipeline-to-plant flange connection was a technical/operational interface owned by the plant contractor (mechanical discipline) with input from operations, whereas aligning the pipeline’s SCADA communications with the plant DCS was an interface between the pipeline contractor and the plant’s instrumentation team. This early cataloguing helped ensure no critical interface was overlooked—late identification of an interface can cause major redesigns and delays [29]. By front-loading interface identification, the project set the stage for proactive management rather than reactive firefighting.
- **Interface Management Plan and Governance:** The project integrated an Interface Management Plan into its Project Execution Plan, following recommended best practices [30][31]. This plan established the governance structure: an interface manager was appointed, interface coordinators were designated within each contractor’s team, and communication protocols were defined. The plan required regular interface meetings (biweekly during design, ramping to weekly/daily during critical phases) and maintained an action log for interface issues. All stakeholders—including the operations reps—had access to this process, reflecting the guidance that brownfield projects demand higher coordination with operations and multiple stakeholders [32][3]. By having a clear governance framework, the Amaken project ensured accountability for each interface and a formal channel for resolving issues.

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- **Centralized Interface Register:** A key tool used was an Interface Register, essentially a live database of all interfaces and their status. This register logged interface descriptions, responsible parties on each side, required deliverables, due dates, and current status (open, closed, pending actions) [33]. For instance, one entry might be "Interface #M-15: 24" pipeline tie-in at Zahran inlet—Responsibility: Pipeline EPC (up to valve) / Plant Contractor (after valve); Deliverables: welding procedure, hydrotest certificate, as-built isometric; Status: Ready for execution (all pre-shutdown tasks done)." The register was kept online for transparency and was referenced in meetings to track progress. A centralized interface register provides a single source of truth, giving visibility of dependencies and ensuring nothing slips through the cracks [33][34]. In fact, when a new subcontractor was brought in late to replace the insulation vendor, the interface register helped onboard them quickly by highlighting all points where insulation work intersected with others—a real-world benefit noted in similar projects [35].
- **Site Verification and As-Built Data Gathering:** Learning from past brownfield issues, the project put strong emphasis on site validation for all interfaces. Early in detailed design, engineering and construction teams jointly performed site walks to locate each tie-in physically. They marked tie-in locations with tags, took photographs and measurements, and engaged plant personnel to confirm the data [36][37]. Critical interfaces (like the main pipeline cut-in) were even 3D laser scanned, as noted, to verify spatial constraints. This process revealed discrepancies in some legacy drawings, allowing the team to update designs in advance. As a rule, existing drawings were not fully trusted until confirmed on-site, which aligns with recommended brownfield practice to avoid surprises [18][19]. By getting plant operations to sign off on field-verified tie-in data, the team ensured everyone had a shared, accurate understanding of the starting point. This significantly reduced last-minute surprises and was instrumental in planning the shutdown work with confidence.
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- **Interface (Tie-in) Readiness Reviews:** Prior to the plant shutdown, the project held dedicated interface readiness review sessions. In these meetings, each planned interface that would be executed during the outage was reviewed in detail: materials availability, manpower, procedures, risk mitigations, contingency measures, and approval status were checked off. These reviews functioned like a “GO / NO-GO” gate for the shutdown: if any interface was not fully ready (e.g. if a spool piece had not arrived or a weld procedure was not approved), it was flagged and resolved ahead of time. This practice echoes the concept of conducting thorough operational readiness reviews, but specifically for interface tasks that have no room for error once the clock is ticking. As part of this, the team aligned all interface work with the plant’s Turnaround (TAR) schedule, ensuring that tie-in activities were synchronized with the plant’s isolation and depressurization steps. Proper interface management means documenting interfaces and aligning them with plant shutdown schedules to ensure continuity of operations [38]. The Amaken project’s readiness checks were credited with enabling a smooth, on-time startup—no small feat given the complexity of the work.
- **Change Control and Communication Protocols:** Given the flurry of changes common in such projects, the interface management process incorporated strict change control. Any proposed change affecting an interface (design modification, scope addition, schedule shift) required an Interface Change Notice to be communicated to all impacted parties, evaluated for risk, and logged in the interface register. For example, when operations added the bypass line late, the change control ensured the pipeline contractor, plant team, and commissioning team all adjusted their plans (procurement of extra materials, updated drawings, and revised test procedures) in a coordinated way. Controlled change management ensures all stakeholders are informed of modifications at interfaces and necessary adjustments are made [39]. Additionally, a structured communication protocol was in place: interface issues were escalated through the Interface Manager to senior management if not resolved within a set timeframe. This kept pressure on timely resolution and gave interface management real weight in the project’s leadership oversight.
- **Use of Interface Management Tools:** In line with modern project management trends, the Amaken team utilized digital tools to support interface management. A dedicated interface tracker module (part of the project’s management software) was employed to send automatic notifications when interface actions were due or updated. Such tools embody industry best practices, where specialized interface management software allows real-time updates, notifications, and dashboards for better visibility across stakeholders [40]. While the tool itself is not a silver bullet, it enforces discipline—for instance, engineers could not mark an interface as “ready” until they uploaded the necessary documents into the system, creating an audit trail [41]. The interface manager also used visual aids like an interface map—a schematic showing all interface points across the pipeline and plant, color-coded by status—to communicate progress during meetings. This visual management helped highlight any areas lagging behind [42]. By leveraging these tools and techniques, the project enhanced transparency and alignment, embodying a “one team” approach despite multiple contractors.

Lessons Learned from the Amaken Project

The Amaken Gas Trunkline case yields a number of lessons learned for managing EPC interfaces in brownfield projects. These lessons can inform future projects to improve performance:

- **Involve Operations Early and Often:** Integrating the plant operations team from the outset is crucial. Their input on shutdown planning, permitting, tie-in feasibility, and safety is invaluable. Early operations involvement helps optimize turnaround activities and tie-ins [43] and ensures that the project's plans align with operational realities. In Amaken, having operations co-own the interface management plan meant permit requirements and shutdown constraints were identified well in advance. Future projects should establish formal liaisons with Operations and even co-create the interface schedule with them.
- **Develop a Comprehensive Interface Register and Tracking Process:** List and track every interface systematically. A robust interface register (or tracker) brings transparency and accountability [33]—everyone knows who is responsible for what and by when. It also serves as a coordination checklist so nothing is forgotten. The Amaken project showed that a live interface tracker, accessible to all parties, can greatly enhance communication and prevent interface tasks from slipping. Recommendation: Use a centralized database or interface management software to manage this register, especially on complex projects.
- **Use Interface Responsibility Matrices and Agreements:** Clearly define ownership and deliverables at each interface. By using an Interface Responsibility Matrix (IRM) and interface agreements, projects can avoid ambiguity. Each interface should have assigned leads on both sides (e.g., contractor A and contractor B), with agreed boundaries of scope [9]. This was a lifesaver in Amaken whenever disputes arose—the IRM settled “who owes what.” Future projects should include an IRM in their interface management plan and ensure all contractors sign on to it.
- **Conduct Field Verification and As-Built Checks for All Tie-Ins:** Do not rely solely on old drawings or assumptions in a brownfield environment. Field-check everything possible. As seen, even small discrepancies can cause big rework later. The practice of physically walking down tie-in points, tagging them, and updating drawings (and even performing advanced 3D laser scans where appropriate) cannot be skipped [18][19]. One lesson is to allocate sufficient time and budget in early project stages for site surveys and as-built documentation—it pays for itself by preventing late changes. Additionally, consider preparing Tie-In Dossiers for each major connection, containing field data, method statements, risk assessments, and drawings, and get them approved by operations well ahead of execution [44].
- **Plan Shutdowns and Tie-ins Meticulously (Interface Readiness):** When a shutdown is involved, planning for it should start early and be extremely detailed. Use tools like tie-in lists or databases to collate all required tie-ins, noting which can be done live vs. which need shutdown, and plan mitigation accordingly [23][24]. The Amaken project's approach of reducing shutdown scope by leveraging redundancies (A/B circuits, isolatable sections, etc.) is a model to follow [23].

- Also, performing Interface Readiness Reviews (analogous to readiness reviews in commissioning) ensures that by the time you get to the outage, all interface-related actions are complete or ready to go. This includes confirming materials are on-site, personnel are briefed, test packs are ready, and contingency plans exist for each critical interface. The lesson: treat the interface tie-in phase as a mini-project within the project—with its own risk register, readiness checklist, and “plan B” for each item.
- **Implement Rigorous Interface Change Control:** Changes are inevitable, especially in brownfield jobs, but uncontrolled changes at interfaces can wreak havoc. Establish a formal change control process specifically for interface points. As learned, when Amaken enforced that any interface change be logged and communicated to all stakeholders, it significantly reduced confusion and scope creep. The project world is rife with examples where a change in one area that wasn’t conveyed led to mismatched parts or scope gaps [13][14]. A good practice is to integrate interface management with the project’s Management of Change (MOC) system [45], so that any MOC affecting another party triggers an interface review. In summary: no interface change is too small to document—better to over-communicate than to have unanticipated surprises during execution.
- **Maintain Continuous Communication and Coordination:** Regular communication was the glue that held the Amaken interface management together. Weekly interface meetings (with minutes and action item trackers) kept everyone in sync [46]. During critical periods, daily toolbox talks among all interface stakeholders on-site helped iron out last-minute details. The “one team” philosophy—treating the various contractors and operations as a unified team for the project—was reinforced often. This cultural aspect is a lesson often understated: all the processes and tools mean little if the teams don’t communicate openly. Future projects should invest in team integration workshops and clearly set expectations that interface management is a shared responsibility (e.g. include interface KPIs in contractors’ performance metrics). As one industry expert noted, project teams must not attempt to delegate interface management away—it requires active collaboration and oversight by the owner and all contractors [47]. [48].

Recommendations for Future Brownfield EPC Interface Management

Drawing on the Amaken Project experience and industry best practices, here are key recommendations for managing interfaces in future brownfield EPC projects:

- **Establish a Formal Interface Management Framework:** Don’t treat interface management as ad hoc. Develop a formal Interface Management Plan aligned with recognized project management standards (e.g., PMI’s project integration processes) at the project outset [43]. This plan should define how interfaces will be identified, tracked, and controlled, and it should assign clear roles (interface manager or coordinators) and authorities. As guidance, brownfield projects in operating facilities will typically warrant a more robust interface management process than simpler projects [49][50]. Gaining management buy-in for this framework is crucial—leadership must understand that interface management is a critical success factor, not an administrative burden.

- **Use Interface Tracking Tools (Interface Registers/Software):** Implement an interface register as a living tool, and consider specialized software for complex projects. A well-maintained interface register provides a centralized view of all interdependencies and progress [33]. Modern interface management software can integrate with schedules (linking interface milestones to the master schedule) and send notifications, greatly enhancing efficiency [40]. At minimum, use a shared spreadsheet or database accessible to all relevant parties. Ensure that the tool captures essential data: interface description, stakeholders, due dates, status, and related documents. The transparency and audit trail this provides will help avoid interface issues turning into costly claims or disputes.
- **Align Interface Management with Risk Management:** Interfaces, especially in brownfield settings, often coincide with high-risk activities (e.g., hot work, major tie-ins). It's recommended to include interface points in the project's risk register and mitigation plans. For instance, classify interfaces by risk level (high-risk ones being those tied to shutdown or safety-critical systems) and focus extra attention on those. Conduct risk assessments specifically for interface activities—this was done informally in Amaken and could be formalized in the future. By treating interface management as intertwined with risk management, projects can prioritize resources on the most critical interfaces and ensure contingency plans are in place.
- **Leverage Interface Coordination Procedures and Checklists:** Develop standard interface management procedures, including checklists for interface handovers and pre-shutdown readiness. Organizations should create templates for interface documentation—e.g., an “Interface Point Agreement” form or a “Tie-in checklist”—to ensure consistency. A checklist for a tie-in might include verifying isolation points, flushing requirements, scaffold/access provision, NDT plans, post-work testing, documentation of reinstatement, etc. The use of such tools enforces thorough preparation and captures lessons for next time. Industry bodies like CII (Construction Industry Institute) have published best practices in this area, emphasizing that structured interface management can save projects tens of millions by averting rework and delays [51]. Companies should incorporate these learnings into their project execution guidelines.
- **Maintain a Live “Tie-In Database” for Brownfield Works:** For projects involving numerous connections to existing facilities, it is highly recommended to maintain a dedicated tie-in list or database. This should enumerate each tie-in with details such as the tie-in ID, associated P&ID, line/service, method (hot tap or cold cut), required outage or not, target date, responsible lead, and status [52]. [25]. During Amaken, this was done through the interface register; however, future projects might keep a separate focused tie-in schedule that is reviewed daily during turnarounds. This ensures that all departments (process, piping, electrical, instrumentation, etc.) coordinate their tie-ins and that the operations team has full visibility. A tie-in database acts as a subset of interface management geared toward execution in live facilities and helps prevent any tie-in from being missed or improperly prepared. In addition, maintain close coordination with any concurrent projects in the same facility – sharing tie-in lists between projects can avoid conflicts where two projects might otherwise unknowingly plan work on the same equipment or at the same time[53].

- **Empower an Interface Management Team:** Ultimately, tools and plans need capable people to implement them. It is recommended to staff an interface manager or coordinator on large brownfield projects—someone whose sole focus is tracking and facilitating interface resolution across parties. This role proved its value in the Amaken project by providing a single point of contact for interface issues. On future projects, especially those with multiple EPC contracts, consider forming a small Interface Coordination Team including representatives from each major contractor and the owner. This team can drive the interface meetings, maintain the tracker, and proactively chase down pending issues. Interface management should be part of performance reviews and incentive schemes for contractors, underscoring that success is a team effort.
- **Capture and Apply Lessons Learned:** After project completion (and even during execution), capture lessons learned specifically about interface management. Brownfield projects offer rich learning opportunities—whether it’s discovering a more effective way to do a tie-in or a communication practice that could be improved. As done here, documenting what went well and what didn’t in interface coordination will help refine the process for subsequent projects. Over time, an organization can build a repository of interface management lessons and checklists. For instance, if a project encountered a permit delay due to late safety studies, a lesson might be to always include an operations safety review in the interface schedule early on. Feeding these lessons back into training and procedures ensures continuous improvement in managing EPC interfaces.

Conclusion

Managing EPC interfaces in brownfield projects is undoubtedly challenging, but the Amaken Gas Trunkline Project illustrates that with diligent planning, open communication, and the right tools, these challenges can be overcome. In such projects, success is often defined by how well the myriad interfaces are handled—those subtle seams where one scope meets another, or construction meets operation. From permit delays and engineering rework to the pressure of a ticking shutdown clock, interface issues have an outsized impact on brownfield outcomes. The lessons learned from Amaken emphasize early and proactive interface management: identify everything, communicate constantly, verify in the field, and leave nothing to chance when live systems are involved. By adopting robust interface management frameworks—including interface registers, coordination teams, readiness reviews, and change control—future project teams can avoid major pitfalls and deliver brownfield projects safely, on time, and on budget [54][17]. In the end, effective interface management is about fostering collaboration across all parties so that a complex project can function as one cohesive effort. As the Amaken project demonstrated, this not only averts delays and cost overruns but also ensures that new facilities come online smoothly within existing operations, achieving the intended business value with minimal disruption. The investment in interface management is an investment in project success—a lesson hard-earned and now clearly proven in the field.

Reference:

- [1] [2] [30] [31] [43] [45] [49] Project Execution Strategies for Successful EPC Management <https://www.long-intl.com/articles/major-project-execution/>
- [3] [4] [5] [6] [7] [9] [10] [11] [12] [26] [27] [28] [29] [32] [33] [38] [40] [42] [46] [50] Project Interface Management <https://engineersnotebook.com/project-interface-management/>
- [8] [47] [48] Interface Management Case Study: Helps Capital Project Finish on Time <https://www.ascertra.com/video-interface-management-case-study-oil-gas-project-finishes-on-schedule>
- [13] [14] [15] [16] [39] [51] Lessons Learned: Why Projects Succeed with Interface Management <https://www.ascertra.com/blog/learning-from-our-mistakes-why-interface-management-is-vital-to-megaproject-success>
- [17] [22] [54] aist.org <https://www.aist.org/getmedia/65642b54-c6ef-4274-b8af-0da785cf2c49/February-2025-088-094.p>
- [18] [19] [20] [21] [23] [24] [44] [53] Brown Field Engineering <https://projectcontrolsonline.com/blogs/63-myblog/622-brown-field-engineering>
- [25] [36] [37] [52] Tie-In List - Industrial Professionals - Cheresources.com Community <https://www.cheresources.com/invision/topic/9976-tie-in-list/>
- [34] [35] How Interface Management Mitigates Contractor Turnover Risk <https://www.ascertra.com/blog/mitigate-construction-risk-with-interface-management-contractor-turnover>
- [41] [PDF] Interface Management—Project Benefits, Results & Restarts https://www.construction-institute.org/CII/media/Documents/blog-Coreworx_IM_Webinar_July22_FINAL.pdf